SPARKY'S SCIENCE FACTS

How Does Sound Travel?

Sound · Ages 7+

> **Short answer:** Sound is made when something vibrates, or shakes very fast. The vibrations travel through the air to your ears.

Sparky always says: Loud sounds can hurt your ears - never shout into someone's ear or put things inside your ear.

How it works

- When you hit a drum, its skin shakes back and forth and pushes the air, making sound waves.
- The waves travel through the air, and make your eardrum shake. Your brain hears this as sound.
- Sound travels about 340 metres every second in air: that is about 1 kilometre in 3 seconds.
- Sound travels even faster through water and solid things like walls and tables.
- In space there is no air, so there is no sound.

Key words: Vibrate: shake back and forth very fast; Eardrum: the thin skin inside your ear

Common mix-ups

- We see lightning before we hear thunder because light is much faster than sound, not because they happen at different times.
- Sound cannot travel through empty space: space battles in films would really be silent.

Did you know? In a summer thunderstorm on the Highveld, count the seconds from the flash to the thunder and divide by 3. That is how many kilometres away the storm is.

Try it: Put your hand on your throat and hum. Feel the vibration! Then stretch a rubber band over a box and pluck it.

Ask Sparky: Why do I hear an echo in a big hall? Can sound travel through water?